

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: DSA0108
PROGRAMMING USING C++

COURSE CODE: OBJECT ORIENTED

COURSE OUTCOME COVERED

CO1: *Apply object-oriented programming principles and C++ constructs such as data types, operators, and control structures to design and write structured OOP-based programs. (BL3 & BL4)*

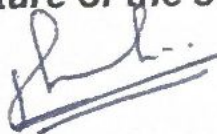
Assessment Tool Weightage: 40%

PSEUDOCODE WRITING

Code of Conduct:

I B.Dhanghina (192572156) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

B. 

Annexure A

QUESTIONS: 10

Total Marks: 100

PSUEDOCODE WRITING

1. A. A teacher wants to identify whether a student roll number is even or odd for seating arrangement. Complete the pseudocode by filling the missing parts to display the correct result. (BL3)

```
START
READ roll
IF roll ____ 2 = 0 THEN
    PRINT "Even"
ELSE
    PRINT "Odd"
END IF
STOP
```

- B. A school wants to display whether a student has passed or failed based on marks. If the mark is 50 or above, display "Pass"; otherwise display "Fail". Fill in the blanks in the pseudocode to implement this logic correctly. (BL3)

```
START
READ mark
IF mark ____ 50 THEN
    PRINT "Pass"
ELSE
    PRINT "Fail"
END IF
STOP
```

- C. A library wants to calculate the fine for late return of books. If the number of late days is 0, no fine is charged; otherwise fine = late days $\times$ 5. Complete the pseudocode by filling the blanks appropriately. (BL3)

```
START
READ days
IF days = 0 THEN
    fine  $\leftarrow$  0
ELSE
    fine  $\leftarrow$  days ____ 5
END IF
PRINT fine
STOP
```

- D. A weather station classifies temperature. If the temperature is above 35, print "Hot Day"; if it is below 20, print "Cool Day"; otherwise print "Pleasant Day". Fill in the missing parts of the pseudocode to classify the temperature correctly. (BL3)


```

START
READ temp
IF temp > 35 THEN
    PRINT "Hot Day"
ELSE IF temp ____ 20 THEN
    PRINT "Cool Day"
ELSE
    PRINT "Pleasant Day"
END IF
STOP

```

E. A sports coach wants to compare the scores of two players and announce the winner. If both scores are equal, print "Match Drawn". Complete the pseudocode by filling the blanks with suitable conditions. (BL3)

```

START
READ score1, score2
IF score1 > score2 THEN
    PRINT "Player 1 Wins"
ELSE IF score2 ____ score1 THEN
    PRINT "Player 2 Wins"
ELSE
    PRINT "Match Drawn"
END IF
STOP

```

2. A small grocery shop gives a discount to customers. If the purchase amount is ₹2000 or more, the customer gets a 10% discount; otherwise, no discount is applied. Construct the pseudocode to read the purchase amount and display the final bill amount. (BL3)
3. A teacher wants to identify whether a student roll number is even or odd for seating arrangement. Write the pseudocode to read a roll number and print whether it is even or odd. (BL3)
4. A tuition center needs to assign grades to students. If marks are 90 and above, assign Grade A; if 75 to 89, assign Grade B; if 50 to 74, assign Grade C; otherwise assign Grade D. Prepare the pseudocode for this grading process. (BL3)
5. A bank wants to verify ATM withdrawal requests. If the requested amount is less than or equal to the available balance, print "Transaction Approved"; otherwise print "Insufficient Balance". Develop the pseudocode for this process. (BL3)
6. A library wants to calculate the fine for late return of books. If the number of late days is 0, no fine is charged. Otherwise, fine = late days × ₹5. Formulate the pseudocode to read the number of late days and display the fine amount. (BL3)
7. A bus pass system checks passenger age. If the age is less than 12, print "Child Fare"; if the age is 60 or above, print "Senior Citizen Fare"; otherwise print "Regular Fare". Design the pseudocode for this system. (BL3)

8. A sports coach wants to compare the scores of two players and announce the winner. If both scores are equal, print "Match Drawn". Write the pseudocode to read two scores and display the result. (BL3)
9. A weather station records temperature. If the temperature is above 35°C, print "Hot Day"; if it is below 20°C, print "Cool Day"; otherwise print "Pleasant Day". Construct the pseudocode for this classification. (BL3)
10. A number-checking system must identify whether an entered number is positive, negative, or zero. Develop the pseudocode for this scenario. (BL3)

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

oops With C++

ASSESSMENT TOOL - 1

SLOT - D

Name : B. Dharshini

Reg. No : 192572156

Course code: DSA0108

Course Name: Object oriented Programming with C++.

PSEUDOCODE WRITING

1, A teacher wants to identify whether a student roll number is even or odd for seating arrangement. Complete the Pseudocode by filling the missing parts to display the correct result.

A. Even or odd

START

READ roll

IF roll MOD 2 = 0 THEN

PRINT "Even"

ELSE

PRINT "Odd"

END IF

STOP

Answer: MOD

B. Pass or Fail

START

READ mark

IF mark $\geq$ 50 THEN

PRINT "Pass"

ELSE

PRINT "Fail"

END IF

STOP

Answer: $\geq$

C, Library Fine

START

READ days

IF days = 0 THEN

fine $\leftarrow$ 0

ELSE

fine $\leftarrow$ days $\times$ 5

END IF

PRINT fine

STOP

Answer: *

D, Temperature Classification

START

READ temp

IF temp $>$ 35 THEN

PRINT "Hot Day"

ELSE IF temp $<$ 20 THEN

PRINT "Cool Day"

ELSE

PRINT "Pleasant Day"

END IF

STOP

Answer: $<$

E. Sports Winner

START

READ Score 1, Score 2

IF Score 1 > Score 2 THEN

PRINT "Player 1 Wins"

ELSE IF Score 2 > Score 1 THEN

PRINT "Player 2 Wins"

ELSE

PRINT "Match Drawn"

ENDIF

STOP

Answer: >

A small grocery shop gives a discount to customers. If the purchase amount is ₹ 2000 or more, the customer gets a 10% discount; otherwise, no discount is applied. Construct the pseudocode to read the purchase amount and displays the final bill amount.

Grocery Shop Discount.

START

READ amount

IF amount $\geq$ 2000 THEN

discount $\leftarrow$ amount * 10/100

ELSE

discount $\leftarrow$ 0

ENDIF

bill $\leftarrow$ amount - discount

PRINT bill

STOP

- 3, A teacher wants to identify whether a student roll number is even or odd for seating arrangement. Write the pseudocode to read a roll number and print whether it is even or odd.

Even or odd Roll Number

START

READ roll

IF roll MOD 2 = 0 THEN

PRINT "Even"

ELSE

PRINT "Odd"

ENDIF

STOP

- 4, A tuition center needs to assign grades to students. If marks are 90 and above, assign grade A; if 75 to 89, assign Grade B; if 50 to 74, assign Grade C; otherwise assign Grade D. Prepare the pseudocode for this grading process.
- Student Grade

START

READ marks

IF marks $\geq$ 90 THEN

PRINT "Grade A"

ELSE IF marks $\geq$ 75 THEN

PRINT "Grade B"

ELSE IF marks $\geq$ 50 THEN

PRINT "Grade C"

ELSE

PRINT "Grade D"

ENDIF

STOP

Q. A bank wants to verify ATM withdrawal requests. If the requested amount is less than or equal to the available balance, print "Transaction Approved", otherwise print "Insufficient Balance". Develop the pseudo code for this.

ATM Withdrawal

START

READ balance, amount

IF amount $\leq$ balance THEN

PRINT "Transaction Approved"

ELSE

PRINT "Insufficient Balance"

ENDIF

STOP

6. A library wants to cal. the fine for late return of books. If the No. of. late days is 0, No fine is charged. otherwise fine = late days $\times$ 5. Develop the pseudocode to read the number of late days & display the final amount.

Library Fine

START

READ days

IF days = 0 THEN

fine $\leftarrow$ 0

ELSE

fine $\leftarrow$ days $\times$ 5

END IF

PRINT fine

STOP

- 7, A bus pass system checks passenger's age. If the age is less than 12, print "child Fare", if the age is 60 or above print "Citizen Fare", otherwise print "Regular Fare". Design the pseudocode for this.

Bus pass System.

START

READ age

IF age < 12 THEN

PRINT "child Fare"

ELSE IF age ≥ 60 THEN

PRINT "Senior Citizen Fare"

ELSE

PRINT "Regular Fare"

ENDIF

STOP

- 8, A sports coach wants to compare the scores of two players and announce the winner. If both scores are equal, print "Match Drawn". Write pseudocode to read two scores and display the result.

Compare Two Scores

START

READ Score1, Score2

IF Score1 $>$ Score2 THEN

PRINT "Player 1 Wins"

ELSE IF Score2 $>$ Score1 THEN

PRINT "Player 2 Wins"

ELSE

PRINT "Match Drawn"

ENDIF

STOP

A Weather station records temperature. If the temperature is above 35°C , print "Hot Day" if it below 20°C , print "Cool Day", otherwise print "Pleasant Day". Construct the pseudo code for this classification.

Weather classification

START

READ temp

IF temp > 35 THEN

PRINT "Hot Day"

ELSE IF temp < 20 THEN

PRINT "Cool Day"

ELSE

PRINT "Pleasant Day"

END IF

STOP

10, A number-checking system must identify whether an entered number is positive, negative or, zero. Develop the pseudocode for this scenario.

POSITIVE, NEGATIVE OR ZERO

START

READ Number

IF number > 0 THEN

PRINT "Positive"

ELSE IF number < 0 THEN

PRINT "Negative"

ELSE

PRINT "Zero"

END IF

STOP